

A Measurement-Grounded Digital Twin for Evaluating Cooperative MARL in Energy-Aware 5G UPF Orchestration

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Abstract—The distribution of 5G User Plane Functions (UPFs) across Mobile Edge Computing (MEC) sites improves service responsiveness but increases the energy footprint of the mobile data plane. Since edge traffic varies substantially across sites and over time, static provisioning often keeps high-performance UPF realizations active even when lower-power alternatives would be sufficient. This paper studies energy-aware UPF orchestration as a cooperative runtime-control problem in which multiple heterogeneous MEC clusters must select between a high-throughput DPDK-based UPF and a lower-power user-space UPF while preserving Quality of Service (QoS) and accounting for switching overheads. We evaluate this problem using a measurement-grounded digital twin calibrated from a physical UPF testbed, incorporating empirical power measurements, QoS surrogate models, and measured activation delays. Within this environment, we compare three reinforcement-learning control architectures: MAPPO with a shared actor and centralized critic, independent per-site PPO agents, and a centralized PPO controller. Results over a held-out test period show that MAPPO consistently outperforms the strongest hysteresis baseline and both PPO alternatives, with statistically significant gains over independent learning. The improvement is most visible in low-load, data-poor clusters, where shared policies transfer useful operating behavior across sites. These findings show that coordination structure is a key factor in energy-efficient UPF orchestration.

Index Terms—5G User Plane Function, energy efficiency, cooperative multi-agent reinforcement learning, MAPPO, digital twin, mobile edge computing, CTDE.

I. INTRODUCTION

Mobile networks are being pushed toward a more distributed architecture. To meet latency requirements and support data-intensive edge services, operators increasingly deploy core-network functions closer to radio access sites and end users. The User Plane Function (UPF) is central in this architecture because it carries packet forwarding, traffic steering, and QoS enforcement in the 5G Core. As UPFs are replicated across Mobile Edge Computing (MEC) sites, their aggregate energy

consumption becomes an important operational cost and a relevant sustainability concern.

The difficulty is that edge traffic is neither uniform nor constant. Some MEC clusters serve dense and persistent demand, while others spend long periods at low load and experience only short peaks. Provisioning every site with a high-throughput data-plane realization is therefore safe but inefficient. It keeps resources active for peak conditions that occur only occasionally. At the same time, using an energy-efficient realization everywhere is unsafe, because it may violate latency or packet-loss constraints when load increases. Energy-aware UPF orchestration must therefore make repeated runtime decisions that balance three competing requirements: reduce power at low load, preserve QoS at high load, and avoid excessive switching between realizations.

This paper focuses on a concrete realization-selection problem. Each MEC site can operate either a DPDK-based kernel-bypass UPF or a user-space UPF. The DPDK realization provides robust high-load performance but maintains a comparatively high idle-power floor because CPU cores remain actively polled. The user-space realization consumes substantially less energy at low load, but its safe operating region is narrower. The resulting control problem is simple to state but difficult to solve in practice. At every 15-minute interval, each site must select the realization that minimizes energy while keeping QoS violations rare, using imperfect short-horizon traffic forecasts and paying the measured cost of changing realization.

A single-site controller can address this problem with thresholds or reinforcement learning. These approaches are useful, but they do not fully capture the operational setting of a fleet of heterogeneous MEC clusters. In a fleet, some sites generate abundant training experience around the relevant switching boundary, whereas low-load sites may rarely observe unsafe states and may not provide enough local evidence for an independent learner to identify when the user-space realization

is safe. In contrast, a controller trained on the combined experience of all sites may learn reusable operating rules, such as when low predicted load justifies the user-space realization, and apply them to sites with limited local data. This motivates a cooperative multi-agent view of UPF orchestration.

The main question addressed in this work is whether cooperation across MEC sites improves energy-aware UPF realization control, and whether the benefit comes from the coordination architecture itself. To answer this question, we compare three architectures under identical measurements, traffic traces, reward definitions, and evaluation protocol. The first is MAPPO, which uses a shared actor for decentralized per-site decisions and a centralized critic through centralized training with decentralized execution (CTDE). The second is an independent per-site PPO ensemble, where each MEC cluster learns its own policy without sharing experience. The third is a centralized PPO policy that observes the global state and outputs the joint action for all sites. This comparison separates the effect of policy sharing and centralized value learning from the broader use of reinforcement learning.

Credible evaluation requires a system model that reflects the actual energy–performance behavior of UPF realizations. For this reason, the study is performed on a measurement-grounded digital twin calibrated from a physical 5G-core testbed. The digital twin is calibrated with measured power, QoS, and switching-delay data from both UPF realizations. As a result, switching is not modeled as a fixed artificial penalty. Instead, its cost depends on the realization being activated and on the offered load, reflecting the behavior observed in the measured system. This allows the evaluation environment to capture the interaction between traffic demand, UPF realization, QoS, and energy consumption more realistically.

The paper makes the following contributions.

- *A cooperative formulation of energy-aware UPF realization control.* We formulate runtime selection between DPDK and user-space UPF realizations across $K = 10$ heterogeneous MEC clusters as a cooperative multi-agent reinforcement-learning problem with binary per-site actions and a shared fleet objective.
- *A measurement-grounded evaluation environment.* We formulate the UPF control problem within a calibrated digital twin derived from physical testbed measurements. The environment combines RAPL-based power measurements collected with Scaphandre, QoS surrogate models for each UPF realization, and empirical activation durations with their associated energy costs. This allows policies to be compared under hardware-grounded energy, QoS, and switching-cost conditions.
- *A controlled comparison of coordination architectures.* We compare MAPPO, an IPPO ensemble, and centralized PPO under the same traffic traces, reward formulation, and switching-cost model. The comparison shows that the coordination architecture has a measurable impact on energy-aware UPF orchestration, with MAPPO achieving the best overall performance among the evaluated learning-based controllers.

The rest of the paper is organized as follows. Section II reviews related work on UPF energy management, reinforcement learning for network control, cooperative MARL, and digital twins. Section III introduces the deployment model, traffic data, and calibrated digital twin. Section IV defines the multi-site control problem and reward. Section V describes the evaluated learning architectures and the evaluation protocol. Section VI presents the experimental results. Section VII concludes the paper.

II. RELATED WORK

A. Energy-aware UPF realization and placement.

Energy reduction in 5G-core deployments has been studied through UPF placement, chaining, scaling, and implementation choices. Placement-oriented formulations optimize where UPFs should be located and how traffic should be routed through them [1]. Complementary measurement studies show that different UPF realizations have markedly different energy–performance profiles: DPDK or VPP-based data planes achieve high packet-processing performance but tend to keep CPU resources active through polling, while user-space implementations exhibit stronger load-proportional behavior [2], [3]. These observations motivate runtime realization selection, where the system switches between a high-performance and a low-power implementation according to offered load.

B. Threshold and hysteresis control.

A common classical approach is to switch realizations using thresholds around an energy–performance crossover. Hysteresis improves over a single threshold by introducing separate upper and lower switching points, thereby reducing oscillation caused by traffic noise or forecast errors. A single-site study reported 10.46% energy savings over static DPDK using polynomial surrogates and an LSTM-based traffic predictor [4]. However, the hysteresis band must be large enough to absorb forecast uncertainty. When the forecast error is large relative to the switching threshold, the lower threshold may collapse to zero, causing the controller to remain in the conservative DPDK mode. In this paper, hysteresis is therefore treated as a strong deterministic baseline: its thresholds are derived from the calibrated twin and its band is swept to identify the best classical operating point.

C. Reinforcement learning for UPF and edge-network control.

Reinforcement learning has been used to adapt UPF deployment and edge-resource management decisions under time-varying demand. Bellin et al. [5] use deep RL to select UPF deployment options at the edge based on measured power behavior. A related single-site formulation [6] models DPDK/USR selection as an MDP evaluated through a measurement-grounded twin and reports energy savings over static DPDK. These works demonstrate that learning-based control can outperform fixed policies, but they focus on one site or treat UPF options as independent choices. The present work extends this direction to a fleet of heterogeneous MEC clusters and asks whether shared learning across sites improves energy-aware realization selection.

D. Cooperative MARL architectures.

Cooperative multi-agent reinforcement learning has been applied to network slicing, radio-resource allocation, traffic steering, and energy-aware radio access control [7], [8]. Value-decomposition methods such as VDN and QMIX decompose a global value into agent-specific terms [9], [10], but their assumptions and off-policy training style are not always convenient for short-horizon operational control with heterogeneous reward scales. MAPPO has emerged as a robust on-policy actor-critic method for cooperative tasks, combining decentralized actors with a centralized critic [11]. At the same time, independent PPO can be highly competitive in some cooperative settings [12]. This makes architecture-level comparison important. Coordination may help, but centralized learning can also introduce optimization difficulties. Our study compares MAPPO, independent per-site PPO, and centralized PPO under a common UPF-control environment.

E. Measurement-grounded digital twins.

Digital twins provide a controlled way to evaluate network-management policies without repeatedly perturbing a live infrastructure [13]. Their value, however, depends on calibration fidelity. For energy-aware control, the twin must reproduce not only operating power but also QoS degradation and the costs of configuration changes. A companion study develops a broader digital-twin vision for energy-aware UPF orchestration in 6G edge systems [14]. This paper uses a concrete calibrated instance of that idea as an evaluation instrument for cooperative MARL.

III. SYSTEM MODEL AND DIGITAL TWIN

A. Deployment model and control loop

We consider a mobile operator with $K = 10$ MEC clusters serving different geographic regions. Each cluster hosts one UPF service that can be instantiated using either a DPDK-based realization or a user-space realization. The control entity operates in the management-and-orchestration layer and issues lifecycle operations to start or stop the corresponding containers at each site. The central 5G Core control plane remains unchanged. The SMF, AMF, UDM, and PCF continue to operate centrally, while the SMF sends N4/PCF rules to the active UPF realization at each MEC cluster. [Check the flow with Chiara.]

At control step t , with a 15-minute period, each cluster i selects an action $a_{i,t} \in \{0, 1\}$, where 0 denotes the DPDK realization and 1 denotes the user-space realization. The joint action across the fleet belongs to $\{0, 1\}^K$ and therefore grows exponentially with the number of sites. This structure makes a centralized joint policy difficult to train as K increases, while also creating an opportunity for policy sharing across sites. The control environment is implemented as a PettingZoo parallel environment [15] that wraps the calibrated digital twin.

B. Digital twin calibration

For each cluster i , time step t , offered load $\lambda_{i,t}$, and selected realization $a_{i,t}$, the digital twin returns operating

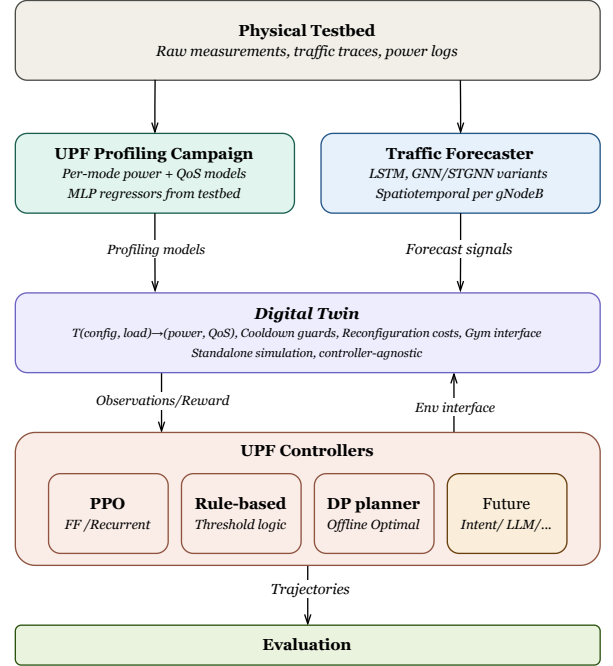


Fig. 1. Measurement-grounded evaluation workflow. The physical UPF testbed is first profiled to fit per-realization power and QoS surrogates. These surrogates, together with traffic traces and short-horizon forecasts, define the digital twin. The same twin is then used as the environment for all evaluated controllers.

power $P_{i,t}^{\text{op}}$, packet delay $d_{i,t}$, packet loss $\ell_{i,t}$, and a binary safety indicator. The twin is calibrated from a physical testbed that includes an SD-Core DPDK UPF, an OpenAirInterface user-space UPF, Keysight LoadCore traffic generation, and RAPL power measurements collected through Scaphandre. The testbed is profiled over various traffic-load levels, creating the measurement basis for the per-realization models.

For each realization m , regression surrogates $p_m(\rho)$ and $q_m(\rho)$ map the normalized load ρ to power and QoS. Each surrogate is chosen by cross-validation: Ridge regression where the response is close to linear, and tree ensembles where saturation makes it nonlinear. Because the twin is the evaluation environment for every controller, we check that these surrogates reproduce the measured behavior on held-out data. The user-space power surrogate reaches $R^2 = 0.99$, and its delay and loss surrogates reach R^2 between 0.97 and 1.00. DPDK power is almost constant across load (standard deviation about 5 mW from 0 to 5 Gbps), and the surrogate reproduces it to within 4 mW, so its low R^2 reflects this near-zero variance rather than a poor fit. The same measurement traces also determine the activation durations of the two realizations: $t_{\text{DPDK}}^{\text{act}} = 24$ s and $t_{\text{USR}}^{\text{act}} = 3.3$ s. These measured values are important because switching to DPDK is substantially more expensive than switching to the user-space realization. The digital twin therefore captures both operating

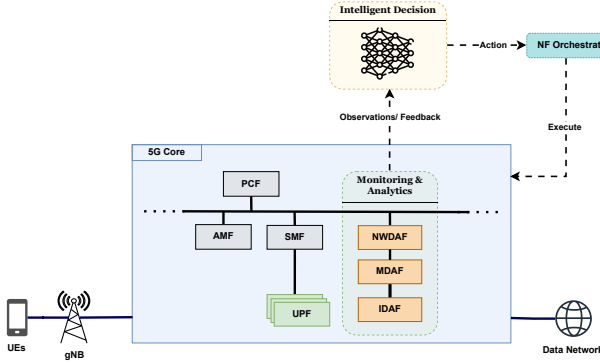


Fig. 2. Deployment model. The controller operates at the management-and-orchestration layer and selects the UPF realization at each MEC site. The 5G Core control plane remains centralized, and the SMF communicates with the active realization through N4/PCF.

[Placeholder]

operation and transition overhead. Live closed-loop validation is left for future work; the present study focuses on controlled, repeatable evaluation inside the calibrated twin.

C. Traffic Traces and Prediction

The traffic process is derived from the NetMob 2023 Lyon dataset, which provides city-scale mobile downlink demand over 100×100 m grid tiles at 15-minute resolution. We use these traces to represent the spatial and temporal variability of urban mobile traffic. Grid-level demand is first mapped to radio-site service areas using a Voronoi-based association constructed from base-station locations. The resulting site-level demand series are then grouped into $K = 10$ heterogeneous MEC clusters, which define the traffic inputs of the digital twin.

To support proactive UPF orchestration, we train a cluster-level spatiotemporal graph neural network, denoted as ClusterSTGNN. The graph nodes correspond to the selected MEC clusters, and edges encode spatial neighborhood relations inherited from the underlying radio-site topology. The model uses a recurrent temporal encoder to capture recent traffic evolution and graph-attention layers to propagate information across neighboring clusters. Given 24 hours of history, the model predicts the traffic load over the next hour. The resulting forecasts are used as demand estimates within the digital twin.

IV. MULTI-SITE CONTROL PROBLEM

A. Observation and Objective

Each agent observes only the local state of its own MEC site, enabling decentralized execution. At decision step t , the observation of agent i contains 14 values: the current offered load $\lambda_{i,t}$, an 8-step load history $\lambda_{i,t-1}, \dots, \lambda_{i,t-8}$, the one-step-ahead traffic forecast $\hat{\lambda}_{i,t+1}$, the previously active UPF realization, the previous QoS score, the previous specific energy (W/Mbps), and a cooldown-progress signal that represents the fraction of the cooldown period that has elapsed since the last switch.

The fleet objective is the discounted sum of per-site rewards over an episode of length $T = 1009$:

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t \sum_{i=1}^K r_{i,t} \right], \quad \gamma = 0.995. \quad (1)$$

The sites are dynamically independent given their selected realizations, but they are coupled through the cooperative training objective.

B. Reward Function

The reward penalizes four quantities: operating energy inefficiency, QoS degradation, switching energy, and rapid oscillation between realizations:

$$r_{i,t} = -(\alpha \text{SEC}_{i,t} + \lambda_{\text{QoS}} \max(0, \tau - Q_{i,t}) + L_{i,t}^{\text{SW}} + L_{i,t}^{\text{CD}}). \quad (2)$$

The first term is based on operating specific energy consumption:

$$\text{SEC}_{i,t} = \frac{P_{i,t}^{\text{op}}}{\max(\lambda_{i,t}, \varepsilon)} \quad [\text{W/Mbps}], \quad (3)$$

where $P_{i,t}^{\text{op}}$ is the operating power of the active UPF realization at time t , excluding activation transients. This term rewards operating modes that carry traffic with lower power per Mbps.

The second term penalizes QoS degradation. The QoS score combines delay and packet-loss behavior:

$$Q_{i,t} = \min(\text{score}_d(d_{i,t}), \text{score}_\ell(\ell_{i,t})). \quad (4)$$

The delay and loss scores decrease linearly from 1, when the metric is within its budget, to 0, when the budget is exceeded by 100%. The budgets are set from the measured testbed limits: 200 μs for delay and 5 packets per interval for loss. The threshold $\tau = 0.9$ defines a safety margin.

The third term accounts for switching energy:

$$L_{i,t}^{\text{SW}} = \lambda_{\text{sw}} E_{i,t}^{\text{sw}}. \quad (5)$$

This penalty is applied only when the active realization changes. The activation energy $E_{i,t}^{\text{sw}}$ is obtained from the digital twin using the measured activation behavior of the target realization. Since activating DPDK takes 24 s while activating the user-space realization takes 3.3 s, switching into DPDK is more expensive.

The final term, $L_{i,t}^{\text{CD}}$, discourages rapid oscillation. It adds a soft surcharge when a site switches again before the four-step cooldown period has elapsed. The reward weights are $\alpha = 100$, $\lambda_{\text{QoS}} = 30$, $\tau = 0.9$, $\lambda_{\text{sw}} = 4.0$, with cooldown period 4 and cooldown cost 0.5.

V. COOPERATIVE MARL CONTROLLERS

A. MAPPO with shared actor and centralized critic

The proposed controller uses MAPPO [11]. Each agent acts through a shared actor network that maps the 14-dimensional local site observation to a binary action distribution over the two UPF realizations. The actor has two hidden layers with 64 units each and 5,250 parameters. Because the same actor

TABLE I
TRAINING CONFIGURATION. VALUES ARE SHARED ACROSS MAPPO,
IPPO, AND CENTRALIZED PPO UNLESS NOTED.

Hyperparameter	Value
Discount γ	0.995
GAE λ	0.9
PPO clip range	0.2
Learning rate	3×10^{-4}
Entropy coefficient	0.05
Rollout length	1024
Minibatch size	256
Epochs per update	10
Reward scaling	0.01
Advantage normalization	per-agent (over time & agents)
Environment steps	200,000
Checkpoint selection	best validation reward
Actor MLP (all)	14–64–64–2 (5,250 par.)
MAPPO critic MLP	140–128–128–10 (35,850 par.)
Seeds	8 (MAPPO, IPPO); 4 (central PPO)

is used by all MEC clusters, the execution policy remains decentralized while the decision rule is learned from experience collected across the fleet.

Training uses a centralized critic. The critic receives the concatenation of the ten local site observations, resulting in a 140-dimensional fleet-level input, and outputs K value estimates. It has two hidden layers with 128 units each and 35,850 parameters. The critic is used only during training to estimate the value function and improve credit assignment across sites.

Training follows PPO clipping [16] with generalized advantage estimation [17], $\lambda = 0.9$, per-agent advantage normalization over both time and agents, and reward scaling of 0.01. The learning rate is 3×10^{-4} , the entropy coefficient is 0.05, the rollout length is 1024, the minibatch size is 256, and each update uses 10 epochs. Each run is trained for 200k environment steps. The validation checkpoint with the best performance is selected for testing, which avoids late-stage policy degradation observed in some seeds.

B. Independent PPO ensemble

The IPPO reference trains ten independent PPO policies [12], one for each MEC cluster. Each policy receives the same 14-dimensional local observation used by the MAPPO actor, but learns only from its own cluster. The ensemble is implemented with Stable-Baselines3 [18] and uses the same PPO hyperparameters as MAPPO where applicable. During evaluation, the ten policies are executed in parallel. This baseline tests whether the benefit of MAPPO comes from shared learning and centralized value estimation rather than from PPO itself.

C. Centralized PPO

The centralized reference uses a single PPO policy that observes the full 140-dimensional fleet state and outputs a joint action using a $\text{MultiDiscrete}([2]^K)$ action space. It

receives the same reward and training budget as the other learning-based controllers. This baseline tests whether a fully centralized policy can solve the joint problem directly, without parameter sharing or decentralized execution.

D. Classical and static baselines

The learned controllers are compared against non-learning references evaluated on the same twin, traffic, and reward.

Always-DPDK keeps every site in the high-performance realization at all times, it is the safe but inefficient static reference that current provisioning corresponds to.

Optimized hysteresis. A per-site, decentralized two-threshold controller that acts on the one-step forecast $\hat{\lambda}_{i,t+1}$, exactly like the RL actors. Its operating points are read from the twin rather than hand-tuned. A load sweep places the energy break-even at 91 Mbps and the USR QoS limit at 149 Mbps, so the upper threshold is their minimum less a 10 Mbps safety margin, $t_{\text{up}} = 81$ Mbps. A site in USR switches to DPDK once $\hat{\lambda}_{i,t+1} \geq t_{\text{up}}$, and a site in DPDK switches to USR only once $\hat{\lambda}_{i,t+1} \leq t_{\text{down}} = t_{\text{up}} - b$; the band b between the two thresholds suppresses oscillation, and after any switch the realization is locked for a four-step cooldown. We call the controller *optimized* for two reasons. First, its thresholds follow from measured twin physics rather than the manual analysis used in our earlier work [4]. Second, we sweep b and report the best-performing operating point ($b = 50$ Mbps). Together these give the classical family its strongest form, so the learned controllers are measured against a well-tuned competitor.

E. Evaluation protocol

All trained controllers are evaluated on the held-out test slice using deterministic actions and a fixed reset seed. MAPPO and IPPO are trained over eight seeds and centralized PPO is trained over four seeds because its failure mode is consistent and large in magnitude. The reported metrics are total reward over all clusters and time steps, total fleet energy in watt-hours, QoS-violation rate, user-space usage share, and switch count.

VI. RESULTS

A. Overall comparison

Table II summarizes the test-slice results. MAPPO obtains the best reward among all evaluated controllers. Relative to the strongest hysteresis configuration, it improves reward by 9.7%. Relative to always-DPDK, it improves reward by 21.5%, and relative to centralized PPO by 33.8%. These values form the 10–34% improvement range reported in the abstract. MAPPO also has the lowest energy among the trained policies, with 1966 Wh compared with 1973 Wh for IPPO and 2071 Wh for always-DPDK. The energy difference between MAPPO and IPPO is small, but MAPPO achieves better reward because its user-space usage is placed more safely, leading to fewer QoS penalties. MAPPO is also the most active controller (Table II), yet its switches are well placed: at a comparable user-space share it attains the lowest energy and QoS-violation rate among the learned controllers, so the additional switching is repaid

TABLE II

TEST-SLICE COMPARISON ($K = 10$, 1009-STEP EPISODE). MAPPO AND IPPO ARE REPORTED OVER 8 SEEDS, CENTRALIZED PPO OVER 4 SEEDS. REWARD IS THE UNWEIGHTED SUM OVER CLUSTERS AND TIME STEPS; QoSV IS THE QoS-VIOLATION RATE; SW/D IS REALIZATION SWITCHES PER CLUSTER PER DAY.

Controller	Reward n	Wh	QoSv%	USR%	Sw/d	
MAPPO (ours)	-5655 ± 98	8	1966	0.54	8.1	2.92
IPPO ensemble	-5812 ± 110	8	1973	0.57	8.2	1.99
Centralized PPO	-8542 ± 1744	4	2066	1.18	4.3	0.44
Hysteresis ($b=50$)	-6261	—	1995	0.68	7.5	1.06
Always-DPDK	-7201	—	2071	0.38	0.0	0.00

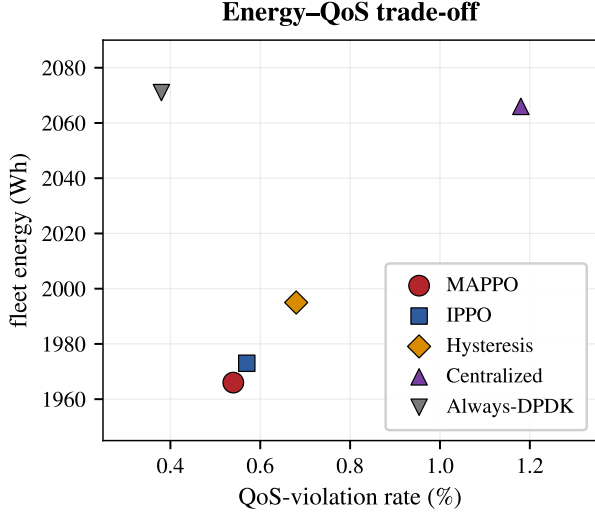


Fig. 3. Energy-QoS trade-off on the test slice (lower-left is better). MAPPO dominates the IPPO ensemble, hysteresis, and centralized PPO on both axes; only static always-DPDK is non-dominated, trading $\sim 5\%$ more energy for marginally fewer violations.

in reward. Figure 3 recasts this comparison as an energy-QoS trade-off: MAPPO occupies the efficient corner, dominating the IPPO ensemble, hysteresis, and centralized PPO on both axes, while static always-DPDK is the only non-dominated alternative—avoiding violations entirely but at a $\sim 5\%$ energy premium.

B. Significance of the MAPPO-IPPO difference

Although the MAPPO advantage over IPPO is smaller than its advantage over hysteresis or centralized PPO, it is statistically robust. A paired bootstrap over the eight seed-matched MAPPO-minus-IPPO reward differences, using 10,000 resamples, gives a mean gain of 157 reward units, a 95% confidence interval of $[+39, +271]$, and a one-sided p -value of 0.004. Seven of the eight paired seeds favor MAPPO. The only IPPO win occurs at seed 23, where MAPPO selects the user-space realization more aggressively and incurs a slightly higher QoS violation rate. The bootstrap distribution and per-seed comparison are shown in Fig. 4.

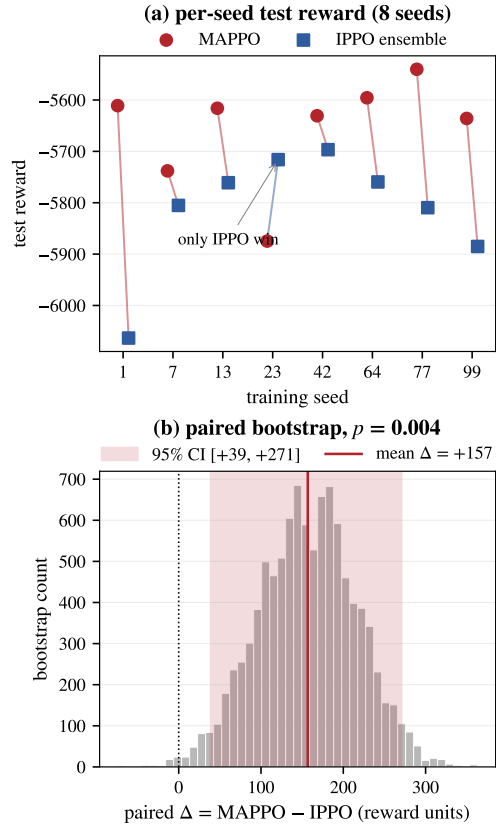


Fig. 4. Seed-paired comparison between MAPPO and IPPO. MAPPO outperforms IPPO on seven of eight seeds. The paired-bootstrap distribution of the mean reward difference excludes zero, with 95% CI $[+39, +271]$ and $p = 0.004$.

C. Per-cluster behavior and transfer across sites

The per-cluster analysis in Fig. 5 explains where the MAPPO gain originates. MAPPO outperforms IPPO on seven of the ten clusters, matches it on a high-load DPDK-bound cluster, and loses slightly on two mid-load clusters. The strongest gains occur on low- and mid-load clusters, including c5, c0, c7, and c4. These are precisely the sites where the user-space realization can be beneficial but where an independent local learner may not observe enough informative transitions to confidently use it.

This pattern is consistent with cross-site transfer through the shared actor. During training, MAPPO observes low-load operating conditions across several clusters and learns a common decision rule for safe user-space usage. That rule can then be applied at sites whose own local trace contains less evidence. In contrast, IPPO must learn each site independently and is therefore more likely to stay close to the conservative DPDK behavior on data-poor clusters. On the highest-load clusters, where DPDK is almost always required, MAPPO correctly avoids excessive user-space usage.

D. Failure mode of centralized PPO

Centralized PPO performs substantially worse than all other controllers. Its average reward is -8542 , with high seed-to-

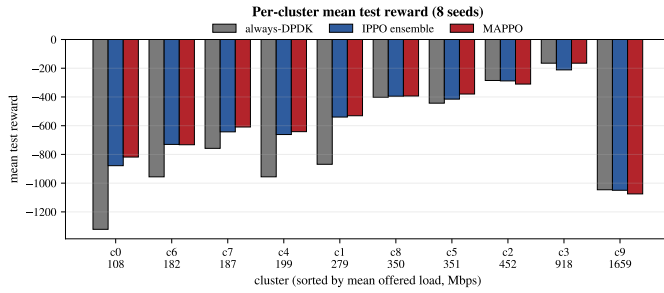


Fig. 5. Per-cluster mean test reward for always-DPDK, IPPO, and MAPPO, sorted by mean offered load. MAPPO’s advantage is concentrated on low- and mid-load clusters, where shared learning helps identify safe user-space operation.

seed variability. The main failure mode is unsafe assignment of the user-space realization to one or two clusters for long periods. In some seeds, individual clusters experience very high QoS-violation rates because the centralized policy does not learn the per-site structure of the problem within the available budget.

This result is important because it shows that more centralized observation does not automatically produce better control. The joint state has 140 dimensions and the joint binary action space grows as 2^K . A shared decentralized actor, by contrast, solves the same local 14-to-2 mapping at each site while using a centralized critic only during training. The comparison therefore supports the main architectural conclusion: cooperative structure must be introduced in a way that preserves the local decision structure of the UPF-control problem.

VII. CONCLUSION

This paper studied energy-aware UPF realization selection across a heterogeneous fleet of MEC clusters. The problem arises because DPDK-based UPFs provide strong high-load performance but carry a high idle-power cost, while user-space UPFs are more energy-efficient at low load but have a narrower safe operating range. Runtime selection between these realizations can reduce energy consumption, but it must be robust to traffic variation, forecast error, QoS constraints, and measured switching overheads.

We formulated the fleet-level problem as cooperative multi-agent reinforcement learning and evaluated it using a measurement-grounded digital twin calibrated from a physical UPF testbed. The twin incorporates per-realization power and QoS surrogates and measured activation times, allowing switching energy to be included directly in specific energy consumption. Under this evaluation setup, MAPPO with a shared actor and centralized critic outperformed the strongest swept hysteresis baseline, an IPPO ensemble, and centralized PPO by 10–34% in reward. The MAPPO advantage over IPPO was statistically significant and was concentrated on low-load clusters, suggesting that shared policies transfer useful behavior from data-rich to data-poor sites.

The results show that coordination architecture is central to energy-aware UPF orchestration. A fully centralized PPO

policy fails to exploit the structure of the problem, whereas a shared decentralized actor combined with centralized training provides both transfer and scalability. Future work will evaluate zero-shot transfer to unseen MEC clusters, incorporate shared-resource coupling among sites, and validate the learned policies in a live closed-loop UPF deployment.

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The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

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